



Artificial Idiots

The promise of AI has been broken too many times. Will we ever get smart computers?

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If you've heard of Isaac Asimov, the Positronic Brain and the Three Laws of Robotics, you're probably as disappointed as we are with the current state of Artificial Intelligence (AI). Sure, computers have evolved from machines the size of buildings to chips the size of your thumb! We're not saying that computers haven't become smarter; we're just asking, by how much?

An AI pioneer called Herbert Simon said in 1965 that machines would, by 1985, be capable of doing any work a man can do. It's 2006 now, and far from "doing any work a man can do," machines can't even use a broom to sweep a room! OK, there are robotic vacuum cleaners, but those just avoid collisions and go around the room—can you even imagine a robot asking you, *in English*, "Should I sweep out that corner as well?" (Because it can see that the corner isn't *all that dirty*, and that *more energy might unnecessarily be expended* doing it—since it *knows from experience* that *you* don't demand that every nook and cranny be *perfectly clean*, because you'd *rather* get the job done quicker?)

We want to see AI put to practical use. Unfortunately, there's hardly any of that happening in the fields we'd like to see it happen in. Sure, there are "expert systems" in the field of medicine and others, and robots that use AI to navigate, but we want to see it in our everyday lives!

But Everyone Claims To Have A Solution and that's one of the prob-

lems plaguing AI. There are so many brilliant minds involved in so much varied research, and no-one seems to be able to arrive at a consensus about what's needed. A robot here is coded to navigate a maze, a chat-bot there is coded to behave like a doctor!

They're all specialised! What we need—or rather, what we want—is a generalised artificial intelligence, something that behaves the way we do: employing common sense, realising when something is repeating, and so on. In fact, common sense and pattern matching—two of the biggest AI problems—just haven't been solved in all these decades! And no, hard-coding common sense by using world knowledge just doesn't cut it.

Imitating human intelligence is another issue: brain modelling apart, there's too much of that going on—when will researchers get to developing an entirely new kind of intelligence? Why does intelligence have to be *human*?

Thinking Machines

We've seen supercomputers beat grandmasters in chess games, but even those instances were riddled with controversy—with the humans claiming that the computer was specifically programmed to beat him! We still haven't seen a computer win a chess tournament, beating many different grandmasters.

But that's just chess! After all, it is a game of logic and mathematics, which come naturally to computers. What we really want to see is a computer that can answer such questions as "Three of us need to enter a room; how many doors are needed?" If a computer or robot can answer these questions using common sense—without specifically being coded to do so—then we will have achieved true artificial intelligence, the *I, Robot* type.

The Current Scene

As of now, there's really nothing new that one would want to write home about. Like we said at the outset, computers and software have become more *knowledgeable*, but not more intelligent.

We need more AI programs to *understand* better. More and more

people are looking to adaptive software and natural language understanding, with logical decision-making in their AI software. This is good, but we're still far from any breakthroughs, and Asimov's Positronic Brain, alas, is just fiction as yet.

A lot of areas of research are proclaiming and promising results soon. However, *soon* is widely believed to be over a decade or two, and the broken promises of the past cause us to take AI timelines with a rather large pinch of salt.

Ai Research (www.a-i.com), for instance, is working on starting off a software as a child, and gradually teaching it and making it smarter. In their words:

"Founded to create true artificial intelligence—making it possible for humans and computers to speak to each other in everyday language—Ai is an international project with a research center near Tel Aviv, Israel.

"The Ai team consists of scientists, researchers, engineers, and language specialists. All work together to nurture Ai's child machine, and to ensure that it will successfully pass the Turing Test within the next 10 years."

("Turing Test" refers to the idea that if a man and a machine can be confused for each other, the machine has passed the Turing Test.)

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If we're only talking about the Turing Test, there might be a few projects that will come close in the next decade. However, making machines as smart as a teenager, even, might take a lot, lot longer.

Right now, we're stuck playing games against moronic AI that is stuck debugging a simple program where you've put an OR instead of an AND! You're also going to be irritated by the same software bugs, again and again, until human research brings about true artificially sentient software, which learns, adapts, and can hold its own against human brains! For now, AI is a huge disappointment, and has been relegated to the stack of technologies that didn't keep their promises. ■

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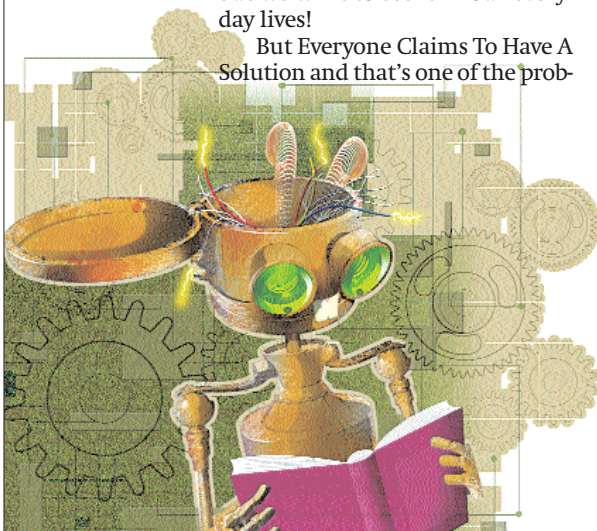


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